

1 Plot Types (Basic Plotting)

`plt.plot(x, y)` - Line plot: Trend / পরিবর্তন দেখাতে `plt.scatter(x, y)` - Scatter plot: Data distribution
`plt.bar(x, height)` - Vertical bar chart: Category comparison `plt.barh(y, width)` - Horizontal bar chart: Category comparison / long category names
`plt.hist(data, bins)` - Histogram: Data distribution `plt.boxplot(data)` - Box plot: Spread & outliers
`plt.pie(data, labels)` - Pie chart: Proportion / share `plt.stackplot(x, y 1, y 2)` - Stacked area plot
`plt.step(x, y)` - Step plot `plt.fill_between(x, y 1, y 2)` - Fill area between lines

2 Figure & Subplot Management

`plt.figure(figsize=(w,h))` - Figure size `plt.subplot(nrows, ncols, index)` - Single subplot `plt.subplots(nrows, ncols)` - Multiple subplots
`fig.add_subplot()` - Add subplot to figure `fig.add_axes()` - Add axes with position
`plt.tight_layout()` - Adjust spacing between subplots

3 Labels, Titles & Legends

`plt.title('Title')` - Plot title `plt.xlabel('X-axis')` - X-axis label `plt.ylabel('Y-axis')` - Y-axis label `plt.xticks()/plt.yticks()` - Custom ticks
`plt.legend()` - Show legend `plt.text(x, y, 'text')` - Add text `plt.annotate('text', xy, xy2, arrowprops)` - Annotation with arrow

4 Colors, Markers & Line Styles

`color='r'/'b'/'g'` - Line / marker color `marker='o','s','^'` - Marker type `linestyle='-', '--', ':'` - Line style
`alpha=0.5` - Transparency `linewidth=2` - Line width `edgecolor/facecolor` - Bars / boxes / patches

5 Axes & Limits

`plt.xlim(a,b)` - X-axis limit `plt.ylim(a,b)` - Y-axis limit `plt.axis('equal')` - Equal scaling `plt.axis('off')` - Hide axes
`plt.gca()` - Get current axes `plt.gcf()` - Get current figure

6 Grid, Ticks & Scale

`plt.grid(True)` - Show grid `plt.minorticks_on()` - Show minor ticks `plt.tick_params()` - Customize ticks
`plt.xscale('linear/log')` - X-axis scale `plt.yscale('linear/log')` - Y-axis scale

7 Saving & Displaying

`plt.show()` - Display plot `plt.savefig('file.png', dpi=300, bbox_inches='tight')` - Save plot `plt.close()` - Close figure
`plt.clf()` - Clear figure `plt.cla()` - Clear axes

8 Advanced Methods

`plt.errorbar(x,y,yerr=...)` - Plot with error bars `plt.fill_between(x, y1, y2)` - Fill area between lines
`plt.stackplot(x, y1, y2)` - Stacked area plot `plt.hexbin(x, y)` - 2D histogram `plt.imshow(image, cmap=...)` - Show image / heatmap
`plt.contour(X,Y,Z)` - Contour lines `plt.contourf(X,Y,Z)` - Filled contour `plt.quiver(X,Y,U,V)` - Vector field plot
`plt.streamplot(X,Y,U,V)` - Flow lines

✓ Quick Tips

- সব সময় `plt.show()` শেষে ব্যবহার
- Styles, colors, markers, alpha দিয়ে plot readable করা যায়
- Subplots ও axes objects ব্যবহার করলে complex visualization সহজ হয়
- Error bars, `fill_between`, `stackplot` ইত্যাদি Advanced plotting এ ব্যবহার করা যায়